



AUGUST/RESIT EXAMINATIONS 2024/2025

MODULE: CSC1028 - Probability & Statistics

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
DS	BSc Data Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,2

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 25]**

For each of these questions **explain your answers** i.e. show your reasoning.

Q 1(a)**[14 Marks]**

A well-shuffled deck of cards contains 52 cards, comprised of 4 suits: 13 Diamonds, 13 Spades, 13 Clubs, and 13 Hearts. Within each suit, each card is labelled Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, and King.

Suppose we were to sample **with replacement** and draw N cards. What is the probability that:

- I. the first card is an Ace or a Club?
- II. the second card is an Ace given that the first card was not an Ace?
- III. the first card is an Ace, the second card is a Spade, and the third card is a Queen?
- IV. the first, third and fifth card are the same suit? e.g. all Diamonds or Spades or Hearts or Clubs.
- V. if we draw 5 cards, that at least two cards drawn are the same?

Q 1(b)**[11 Marks]**

A box contains 25 circuits. We are told 20% of the circuits in the box are defective.

- I. If 3 circuits are taken from the box without replacement, what is the probability all 3 circuits are defective?
- II. If 3 circuits are taken from the box without replacement, what is the probability that 2 or more of these circuits are defective?
- III. If the box of circuits was infinite in size, how would this change how we calculate part (I) and part (II) above? Explain your answer.

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]**

For each of these questions **explain your answers** i.e. show your reasoning.

Q 2(a)**[21 Marks]**

Five delivery companies (labelled A, B, C, D and E) are used by an organisation to send packages that contain documents. The table below indicates the total number of deliveries that were on time or late, as well as the fee charged by each delivery company to deliver a package. The fee is still charged regardless of whether the package was on-time or late.

Delivery company (ID)	A	B	C	D	E
Delivery fee per package (€)	1	2	4	6	1
Total number of on-time deliveries	14	9	3	4	7
Total number of late deliveries	5	2	1	2	3

- I. What are the mean and variance of the expected delivery fee for the organisation?
- II. What is the probability that delivery company C, D or E was used, if we are told the delivery fee is less than €5?
- III. If we are told either company A or B was used to send a package, what is the probability that the package was late?
- IV. If two separate deliveries were made to the same recipient on a particular day, what is the probability two different delivery companies were used? (note: *you may assume the delivery companies were chosen independently with probabilities based on past delivery frequency.*)

Q 2(b)**[4 Marks]**

The organisation uses the following formula to calculate the cost of preparing a package for delivery based upon the number of documents it contains:

$$\text{Cost} = 2X + 3$$

Where **X** = the number of documents in the package

We are told the average number of documents in a package is 4, with a variance of 4. What is the expected variance of the cost to prepare a package for delivery? i.e. find $V(2X + 3)$

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]**

A programmer writes programs in one of either Java, Fortran or Kotlin, where some of these programs contain errors. We are told 50% of all programs are written in Java, 20% in Fortran and 30% in Kotlin.

Examining only the programs that produce errors, we find 40% are written in Java, 10% in Fortran, and 50% in Kotlin.

We are also told that 20% of the programs written in Java contain errors.

For each of these questions **explain your answers** i.e., show your reasoning.

Q 3(a) **[3 Marks]**

If we are told a program is written in Java, what is the probability that it contains no errors?

Q 3(b) **[4 Marks]**

What is the probability a program is written in Kotlin, if we are told that the program is not written in Java?

Q 3(c) **[7 Marks]**

What is the probability if a program is chosen at random that it contains errors?

Q 3(d) **[6 Marks]**

What is the probability that a Fortran program contains no errors?

Q 3(e) **[5 Marks]**

What is the probability a program is written in Fortran and contains errors?

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 25]**

In a garage open 24 hours a day, an average of 4 cars arrive each day. Of these cars, 75% are for servicing, while the remaining 25% need repairs. Cars are equally likely to arrive at any time of day or on any day of the week.

For each of these questions **explain your answers** i.e., show your reasoning.

Q 4(a) **[2 Marks]**

What is the average number of hours between the arrival of cars?

Q 4(b) **[4 Marks]**

What is the probability that no cars will arrive in a 24-hour period?

Q 4(c) **[4 Marks]**

What is the probability that exactly 2 cars arrive for servicing on the same day?

Q 4(d) **[4 Marks]**

What is the probability of having to wait less than 5 hours for a car to arrive for repair?

Q 4(e) **[5 Marks]**

For what duration of time (in hours) would the probability of no cars arriving be 0.2231?

Q 4(f) **[6 Marks]**

What is the probability that the same number of cars will arrive at the garage for repairs as for servicing on a particular day? Provide your answer to 2 decimal places.

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Answer any 2 of the following questions i.e. of Q5(a), Q5(b) or Q5(c). There are equal marks for both questions.

For each of these questions **explain your answers** i.e. show your reasoning.

Q 5(a)

[12.5 Marks]

A production line produces devices consisting of four (4) types of components in a series system: C1, C2, C3 and C4. We are told there is a 0.1 probability that C1 is defective, a 0.2 probability that C2 is defective, a 0.3 probability that C3 is defective, and a 0.4 probability that C4 is defective. **note:** *in this series configuration, if a single component is defective, the whole device is defective.*

- I. What is the probability that a device is defective?
- II. What is the probability that the first non-defective device from the production line will be the 3rd device inspected?
- III. If each component is replicated in parallel, what is the probability that the system is defective?

Q 5(b)

[12.5 Marks]

The average hat size is 10 with a standard deviation of 2. You can assume that hat sizes follow a normal distribution.

- I. What is the probability that a randomly selected person has a hat size less than 12?
- II. What is the probability that the hat size of a person is less than 12, if we are told their hat size is greater than 8?
- III. Find the hat size x such that the probability of a randomly selected person having a hat size greater than x is 0.3085

Q 5(c)

[12.5 Marks]

Two players in a game roll a 6-faced die each per round. There are three possible outcomes in each round: both players have the same value (they draw), player A has a higher value than player B (player A wins), or player B has a higher value than player A (player B wins).

- I. What is the probability that the players will not draw for three rounds of the game?
- II. What is the probability that for 1 or more of 5 rounds of the game that the players will draw?
- III. What is the expected mean and variance of the number wins for player A if they play 10 rounds?

[End of Question 5]

APPENDICES

Area under the standard normal curve from $-\infty$ to z



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6338	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

${}^nC_k = \frac{{}^nP_k}{k!} = \frac{n!}{k!(n-k)!}$	
${}^nP_k = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$	
$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$	
$P(A B) = \frac{P(A)P(B A)}{P(B)}$	$P(A \cap B) = \frac{P(A)P(B A)}{P(B)P(A B)}$
$V(X) = \sum_x (x - \mu_x)^2 p(x)$	$E(X) = \sum_x xp(x)$
$P(X = x) = \binom{n}{x} p^x q^{n-x}$	$P(X \leq x) = 1 - q^x$ $P(X = x) = q^{x-1} p$
$P(X = x) = \frac{\binom{M}{x} \binom{L}{n-x}}{\binom{N}{n}}$	$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!}$

[END OF APPENDICES]

[END OF EXAM]